Mission-Grade TT&C Communication System for CubeSats

User & Operations Manual

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1. Overview

This manual describes how to operate the ground segment to command the satellite node and interpret returned telemetry during a simulated pass. It is written for an operator who has not previously interacted with the system, per the task requirement that an independent operator be able to run the demonstration from this document alone.

2. Ground Station Setup and Operating Procedure

1. Power on the satellite node (STM32 + LoRa) and confirm 'Satellite Ready' appears on its debug serial output.

2. Power on the ground station node (STM32 + LoRa + ESP32 bridge). Confirm 'GCS READY' appears on the ESP32 UART line.

3. Connect the ESP32's USB port to the operator PC.

4. Launch the ground control GUI (ground_control_gui.py). Requires Python 3 with the pyserial package installed (pip install pyserial).

5. In the GUI, click Refresh to populate the COM port list, select the ESP32's port, and click Connect. The status indicator should change to CONNECTED.

6. Click PASS START to begin the simulated pass window and arm Doppler compensation tracking on the ground station.

7. Issue telecommands using the LED ON, LED OFF, BLINK, and STATUS buttons as described in Section 3.

8. Monitor the console log and KPI panel for TX/ACK/TIMEOUT events, PER, average latency, Doppler shift, and FEC-corrected bit counts.

9. When finished, click Disconnect, then close the GUI window (telemetry_log.csv is flushed and closed automatically on exit).

3. Telecommand Format Specification

Commands are entered via the GUI and sent as short text tokens over the ESP32-to-GCS UART link; the GCS firmware translates each into an authenticated, FEC-protected wireless frame. The GUI-to-GCS text commands are:

GUI Command	Wire Command Byte	Payload	Effect on Satellite
LED_ON	0x01	0	Illuminates onboard LED, sets state = ON
LED_OFF	0x02	0	Extinguishes onboard LED, sets state = OFF
BLINK:<n>	0x03	n (blink count)	Blinks LED n times, 250 ms on / 250 ms off, then restores prior state

GUI Command	Wire Command Byte	Payload	Effect on Satellite
STATUS	0x10	0	Requests acknowledgment carrying current LED state as telemetry
PASS_START	n/a (local GCS command)	n/a	Resets the ground station's pass timer to t=0 for Doppler compensation

On the wireless link, every command is carried inside the 9-byte logical frame defined in the Implementation Description (SYNC, SRC, DST, SEQ, CMD, DATA, MAC, CRC16), which is then FEC-encoded and interleaved into an 18-byte wire frame before transmission.

4. Telemetry and Response Format

Every command produces one of three outcomes, each logged in the GUI console and archived to telemetry_log.csv:

Console Line	Meaning
TX seq=N attempt=A doppler_hz=D	Ground station transmitted attempt A of command sequence N, with Doppler correction D Hz applied
ACK seq=N latency_ms=L retries=A corrected=C	Satellite acknowledged sequence N after L ms round-trip, on retry attempt A, with C bit errors corrected by FEC
NACK seq=N corrected=C	Satellite received and validated the frame but rejected the command itself (e.g., unrecognized command byte)
TIMEOUT seq=N	No valid ACK/NACK was received for sequence N within all retry attempts
DROPPED: bad CRC/MAC ...	A frame failed integrity or authentication checking after FEC decoding and was discarded
DROPPED: replay seq=N	A frame with a previously-seen sequence number was rejected by the anti-replay window

5. Operational Sequence for a Simulated Satellite Pass

This sequence corresponds to the mandatory Phase 3 closed-loop demonstration described in the task brief:

1. Confirm both nodes are powered and the GUI shows CONNECTED.

2. Click PASS START to simulate acquisition of signal (AOS) and begin Doppler tracking.

3. Send LED_ON. Confirm the satellite's physical LED illuminates and the GUI logs a matching ACK line.

4. Send STATUS. Confirm the ACK's underlying DATA field reflects LED state = ON (visible via the satellite's own serial debug output printing 'STATUS REQUEST').

5. Send BLINK:5. Confirm five visible blink cycles occur on the satellite hardware before it returns to its prior LED state and sends its ACK.

6. Send LED_OFF. Confirm the LED extinguishes and a final ACK is logged.

7. Allow the pass timer to run to completion (10 minutes) or observe Doppler shift in the KPI panel returning toward zero, simulating loss of signal (LOS).

8. Export or archive telemetry_log.csv as the evidentiary record of the demonstration run.

6. Troubleshooting

Symptom	Likely Cause	Action
GUI shows no COM ports	ESP32 not connected or driver not installed	Check USB cable/port; install CP210x or CH340 driver as appropriate for the ESP32 board
Connect succeeds but no console activity	Wrong baud rate or wrong COM port selected	Confirm 115200 baud on all three serial links (ESP32-PC, ESP32-GCS, GCS debug)
Repeated TIMEOUT with no ACK	Out of range, antenna disconnected, or satellite node not powered	Reduce distance, verify antenna connections, confirm satellite 'Satellite Ready' message
Frequent DROPPED: bad CRC/MAC	Link margin too low or FEC overwhelmed by burst interference	Reduce distance/interference; this is expected behaviour at the edge of link margin, not a firmware fault
Doppler KPI stuck at 0 Hz	PASS START was not clicked, or the pass timer already expired	Click PASS START to re-arm the 10-minute compensation window